Experiment-01

Study of different types of LAN, Interconnection Devices & Network Equipment.

Aim

To study and understand the different types of Local Area Networks (LAN), their working principles, interconnection devices, and commonly used network equipment.

Hardware Used:

- Computer / Laptop
- Network Interface Card (NIC)
- Ethernet Cables
- Hub, Switch, Router 1
- Modem
- Wireless Access Point

Software Used:

- Nill

Theory

A computer network is a collection of interconnected devices that can communicate with each other to share data, resources, and services. A Local Area Network (LAN) is a network that covers a small geographical area such as a room, building, or campus.

LANs provide high-speed communication, improved resource utilization, centralized data management, and enhanced reliability. They are widely used in homes, offices, schools, industries, and institutions.

Key differences between LAN & WAN

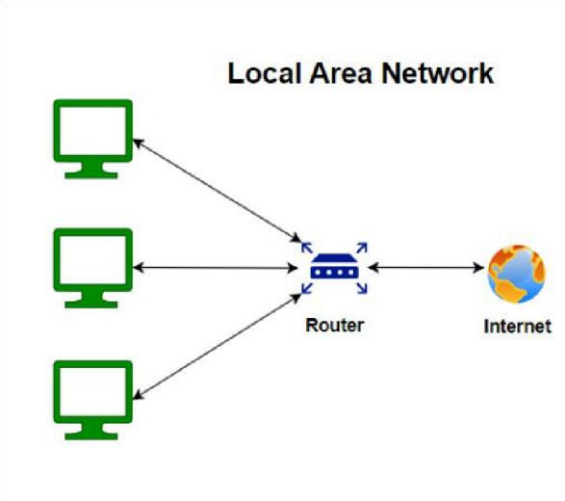
- LAN covers a small geographical area like a home or office, while WAN covers a large area like cities or countries.
- LAN offers high data transfer speed, whereas WAN has comparatively lower speed.

- LAN is inexpensive and easy to maintain, while WAN is costly to set up and manage.
- LAN is usually owned by a single organization, whereas WAN is owned by multiple service providers.
- Examples of LAN include school or office networks, while the Internet is the best example of WAN.

Types of LAN

1. Wired LAN:

A wired LAN uses physical transmission media such as Ethernet cables to connect devices. It offers high data transfer speed, low interference, and better security compared to wireless networks. Wired LANs are commonly used in offices and laboratories.



2. Wireless LAN (WLAN):

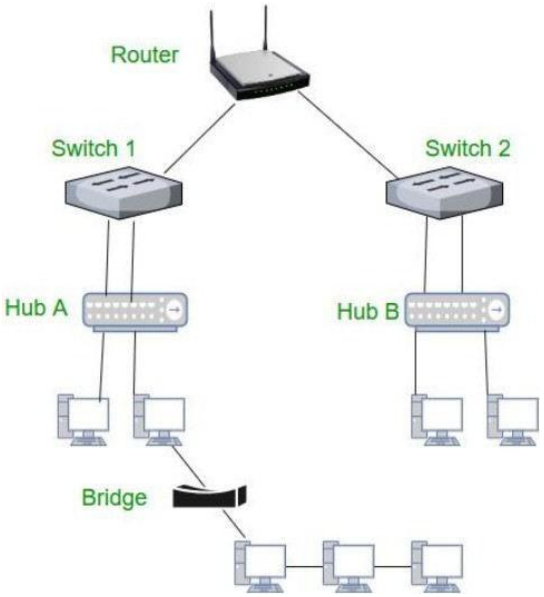
A wireless LAN uses radio waves instead of cables to connect devices. Wireless Access Points (WAPs) allow devices such as laptops, smartphones, and tablets to connect to the network. WLANs provide flexibility and mobility.



Interconnection Devices

Interconnection devices are networking components used to connect multiple devices or networks and control data flow.

- Hub: A hub is a basic networking device that sends data to all connected devices.
- Switch: A switch intelligently forwards data only to the destination device.
- Router: A router connects different networks and determines the best path for data packets.
- Bridge: A bridge divides a network into segments to reduce traffic.
- Repeater: A repeater amplifies signals to extend network distance.



Network Equipment

Network equipment refers to hardware components required for network communication.

- Network Interface Card (NIC): Enables a device to connect to a network.
- Modem: Converts digital signals into analog signals and vice versa.
- Access Point: Allows wireless devices to connect to a wired LAN.
- Cables and Connectors: Provide physical connectivity between devices.

Conclusion

This experiment helped in understanding the structure and working of LANs, the role of interconnection devices, and various network equipment. Such knowledge is essential for designing, installing, and troubleshooting computer networks.

Experiment-02
Study of Various Transmission Media

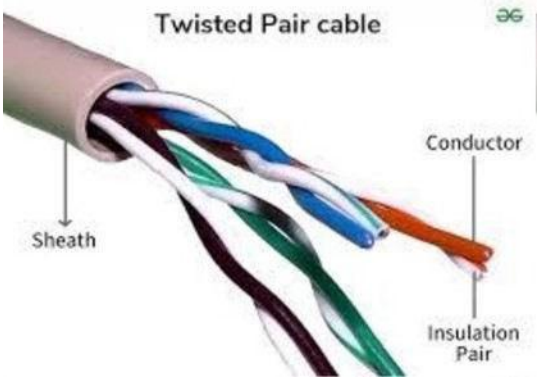
Aim
To study the different types of transmission media used in computer networks and understand their characteristics and applications.

Hardware Used Software Used
- Computer system
- Cisco Packet Tracer (or any network simulation tool)
- Internet access

Theory
Transmission media refers to the physical path through which data travels from one device to another in a computer network. It acts as a communication channel that carries signals between network devices.
Transmission media are broadly classified into Guided (Wired) Media and Unguided (Wireless) Media.

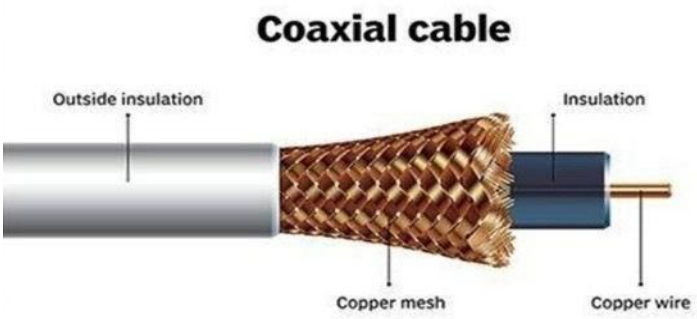
1. Guided Transmission Media:
Guided media use physical cables to transmit data signals from one device to another.

a) Twisted Pair Cable:
Twisted pair cable consists of two insulated copper wires twisted together to reduce electromagnetic interference. It is widely used in telephone networks and Ethernet networks.



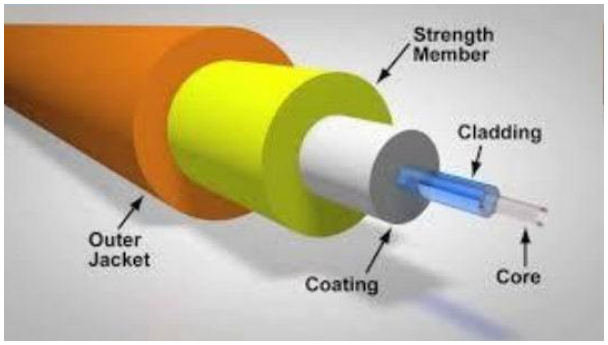
b) Coaxial Cable:

Coaxial cable consists of a central copper conductor surrounded by insulation, a metallic shield, and an outer protective layer. It provides better protection from interference compared to twisted pair cables.



c) Optical Fiber Cable:

Optical fiber cables transmit data in the form of light through thin strands of glass or plastic fibers. They provide very high data transmission speed and are widely used in long-distance communication.

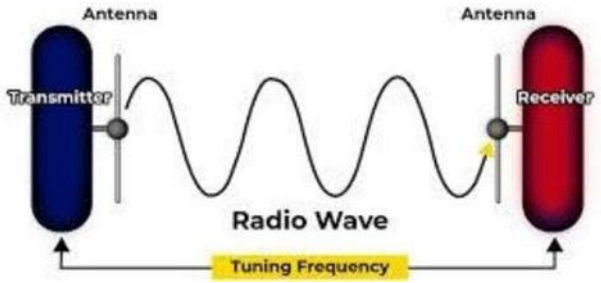


2. Unguided Transmission Media

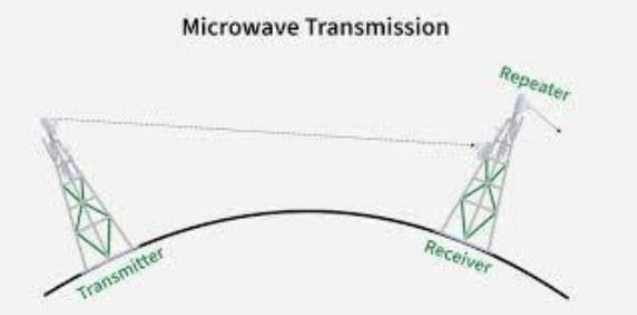
Unguided media transmit signals through air or space without using physical

cables. a) Radio Waves:

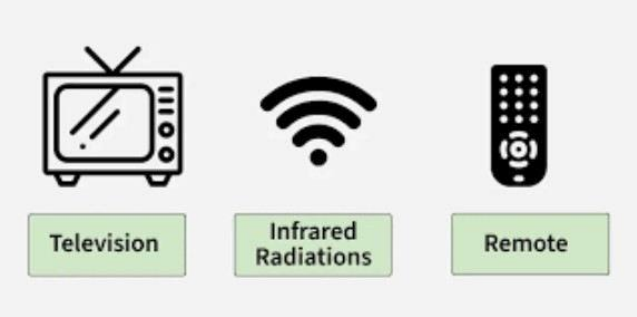
Radio waves are used for wireless communication such as radio broadcasting and wireless networking.



b) Microwaves:
Microwave transmission uses high-frequency signals and is commonly used in satellite communication and long-distance communication links.



c) Infrared:
Infrared transmission uses infrared light waves and is commonly used in shortrange communication such as remote controls and some wireless devices.



Conclusion

Transmission media play a crucial role in computer networking by enabling the transfer of data between devices. Different types of guided and unguided media are used depending on the required speed, distance, cost, and reliability of communication.

Experiment-03

Interpretation and Execution of networking commands like Ping Traceroute and ipconfig etc.

Aim

To study, interpret, and execute basic networking commands such as Ping, Traceroute, and ipconfig for network troubleshooting and configuration.

Hardware Used Software Used

- Computer system (Windows/Linux)
- Command Prompt / Terminal
- Internet connection

Theory

Networking commands are essential tools used to diagnose, troubleshoot, and configure network connections. They help users understand connectivity, routing paths, and system network settings.

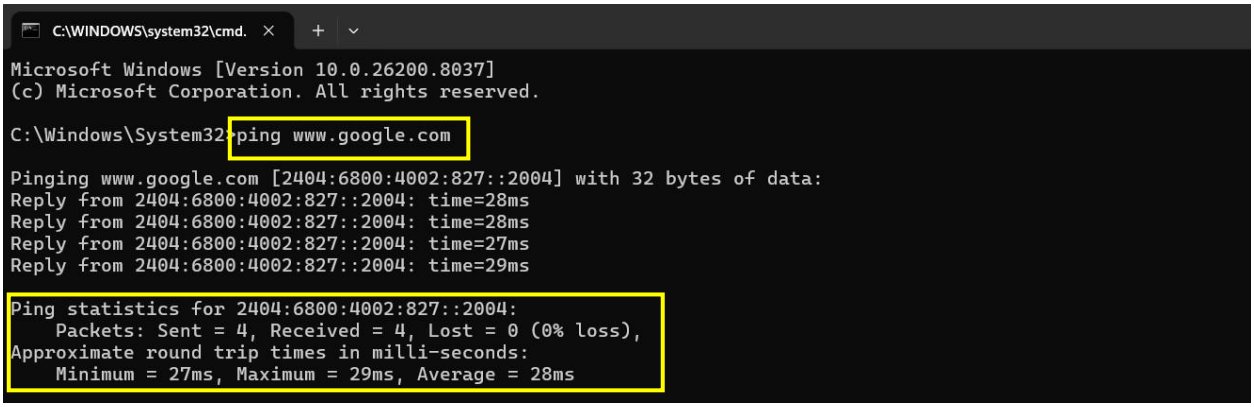
1. **Ping Command:**
   Ping is a network utility used to test the reachability of a host on an IP network using ICMP echo request and reply messages.

Usage:

- Checks whether a device is connected to the network
- Measures round-trip time (RTT)
- Detects packet loss Syntax:

ping <IP address / domain name>

Example:
ping google.com

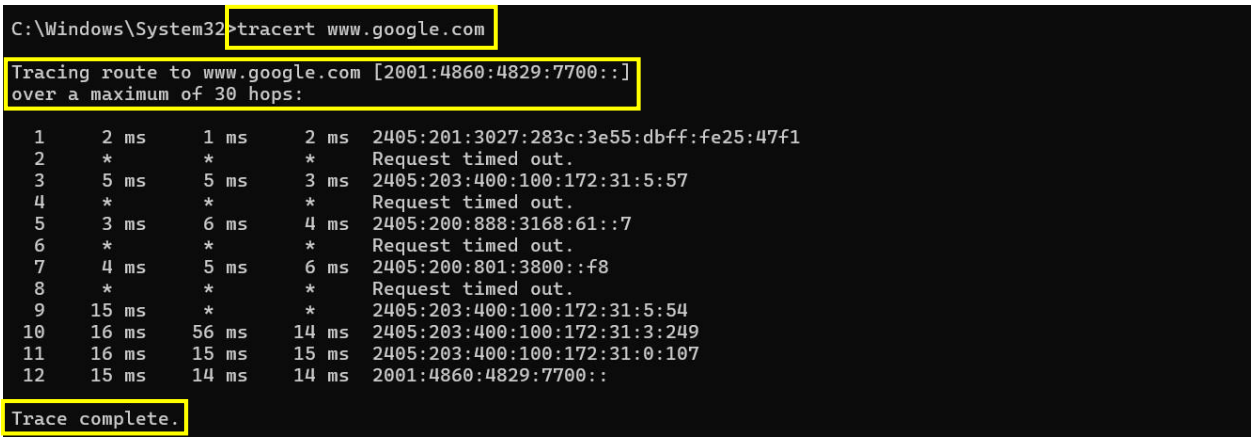


2. **Traceroute Command**
   Traceroute (or tracert in Windows) is used to trace the path taken by data packets from source to destination, showing all intermediate routers.

Usage:

- Identifies routing path
- Detects network delays or failures
- Helps in troubleshooting network congestion Syntax: tracert <IP address / domain name> (Windows) traceroute <IP address / domain name> (Linux)

Example:
tracert google.com



3. **ipconfig Command** ipconfig is used to display and manage the IP configuration of a computer system.

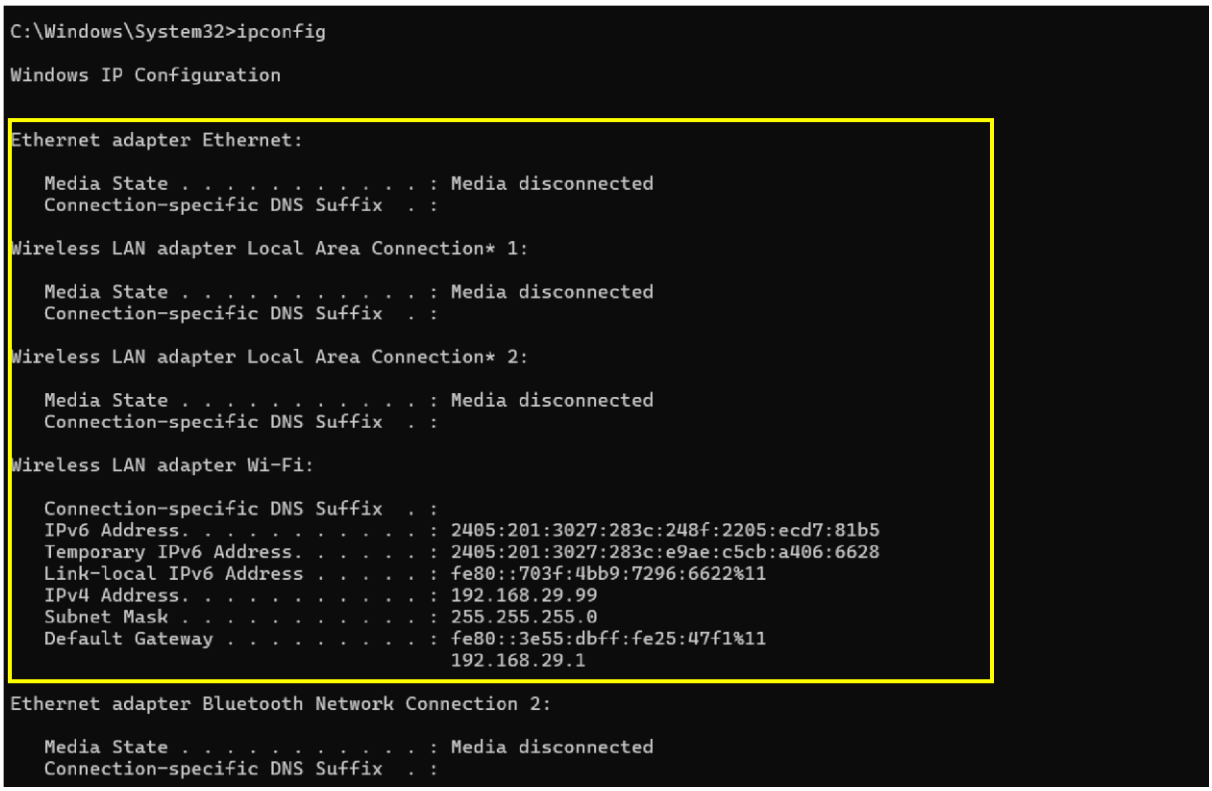
Usage:

- Displays IP address, subnet mask, and default gateway
- Helps in troubleshooting network configuration issues
- Used to release and renew IP addresses Syntax: ipconfig

Common Options:

- ipconfig /all – Detailed configuration
- ipconfig /release – Release IP
- ipconfig /renew – Renew IP

Example:
ipconfig



4. **nslookup Command** nslookup is used to query DNS servers to obtain domain name or IP address mapping.

Usage:

- Finds IP address of a domain
- Troubleshoots DNS issues

Syntax:
nslookup <domain name>

Example: nslookup
google.com



Conclusion

Networking commands like Ping, Traceroute, ipconfig, netstat, and nslookup are essential for analyzing and troubleshooting networks. They provide valuable insights into connectivity, routing, and configuration.

Experiment-04
LAN installations and Configurations.

Aim
To study the installation and configuration of a Local Area Network (LAN) using Cisco Packet Tracer and understand IP addressing and network setup.

Hardware Used
- Computer system
- Ethernet cables (in simulation)
- Switch/Router (in Packet Tracer)

Software Used
- Cisco Packet Tracer

Theory
Introduction to LAN (Local Area Network)

A Local Area Network (LAN) is a network that connects computers and devices within a limited geographical area such as a home, office, or campus. It enables communication, data sharing, and resource sharing using networking devices like switches and routers.

IP Addressing in LAN Configuration

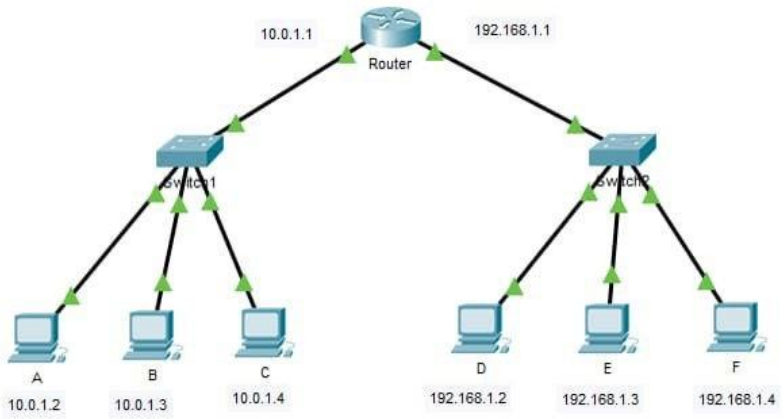
For communication between devices in a LAN, each device must have a unique IP address. IP addresses can be assigned in two ways:

1. Static IP Address
    - Manually assigned to a device
    - Remains constant unless changed manually
    - Used for servers, printers, and important devices
2. Dynamic IP Address
    - Automatically assigned by DHCP (Dynamic Host Configuration Protocol)
    - Used for client devices
    - Easier to manage in large networks

Procedure for LAN Configuration (Using Cisco Packet Tracer)

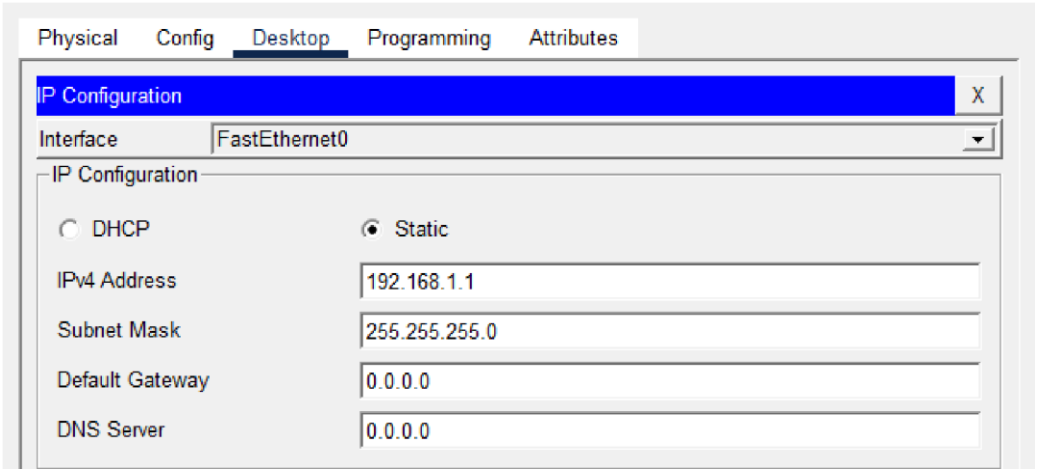
Step 1: Creating Network Topology

- Open Cisco Packet Tracer
- Add 2–3 PCs and one Switch and one router
- Connect PCs to the switch using Copper Straight-Through cables



Step 2: Configuring IP Address on PCs

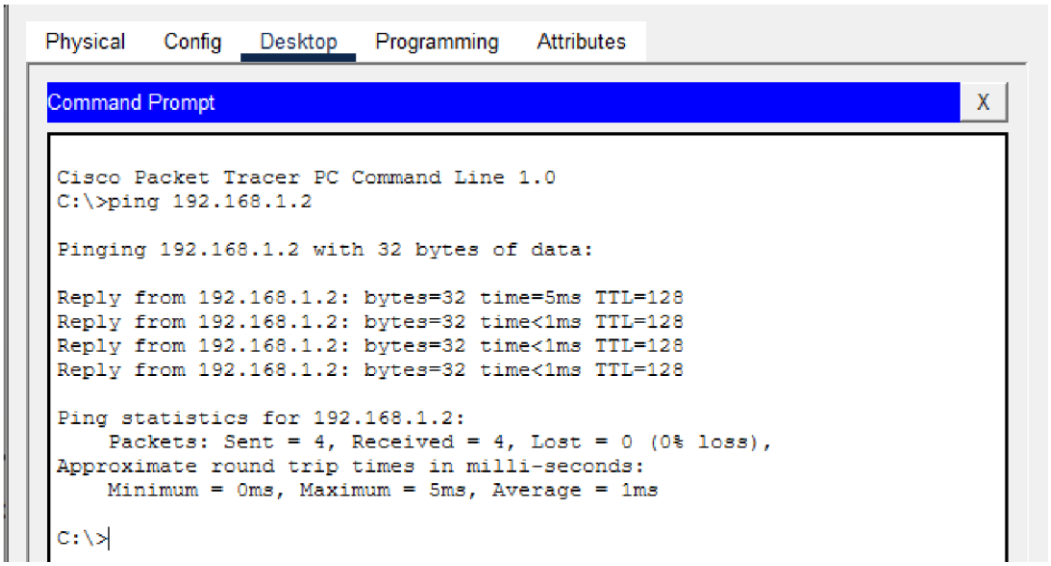
- Click on a PC → Go to *Desktop* tab → Select *IP Configuration*
- Enter the IP Address and Subnet Mask manually Example:
- PC1: 192.168.1.1
- PC2: 192.168.1.2
- Subnet Mask: 255.255.255.0



Step 3: Verifying Connectivity Using Ping

- Click on a PC → Go to *Desktop* → Open *Command Prompt*
- Use ping command to check connection Example:

ping 192.168.1.2



Conclusion

This experiment demonstrated LAN setup and configuration using Cisco Packet Tracer. It included creating a network topology, assigning IP addresses, and verifying connectivity, which are essential for building and managing networks.

Experiment-07
Installation of Cisco Packet Tracer

Aim
To install Cisco Packet Tracer software successfully on a computer system and understand its basic usage for network simulation and learning purposes.

Hardware Used
- Computer or Laptop

Software Used
- Internet Connection
- Cisco Networking Academy Account
- Cisco Packet Tracer Installer Software
- Operating System (Windows / Linux / macOS)

Theory
Cisco Packet Tracer is a network simulation tool developed by Cisco Systems. It enables users to design, configure, and test network topologies virtually. The software provides a graphical interface for routers, switches, and end devices, allowing learners to understand networking concepts without physical hardware.

Installation Screenshots
Figure 1: Cisco Packet Tracer Download Page

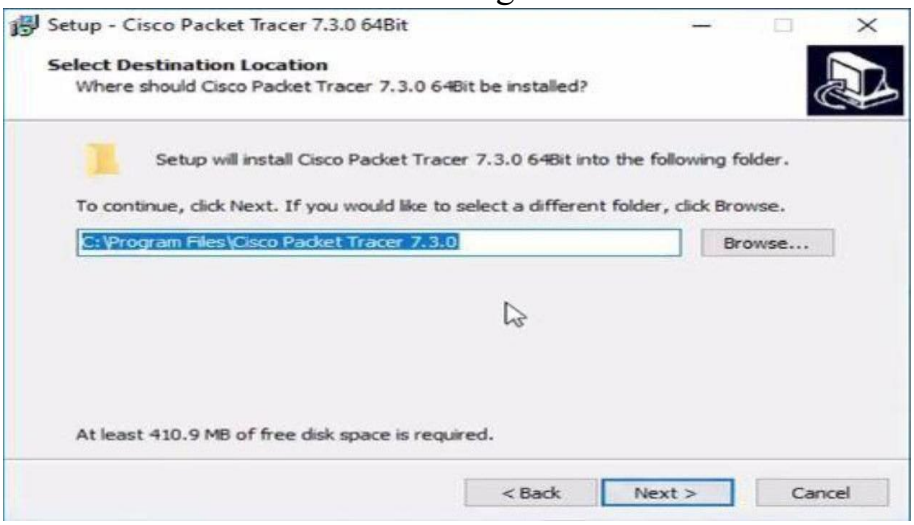


Figure 2: Cisco Packet Tracer Installation Wizard

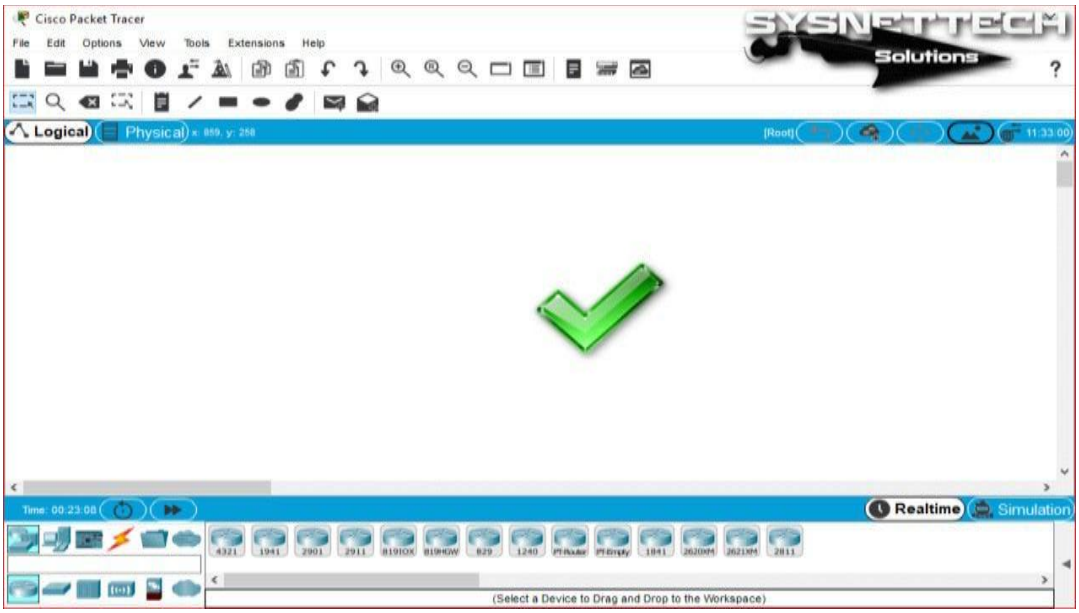
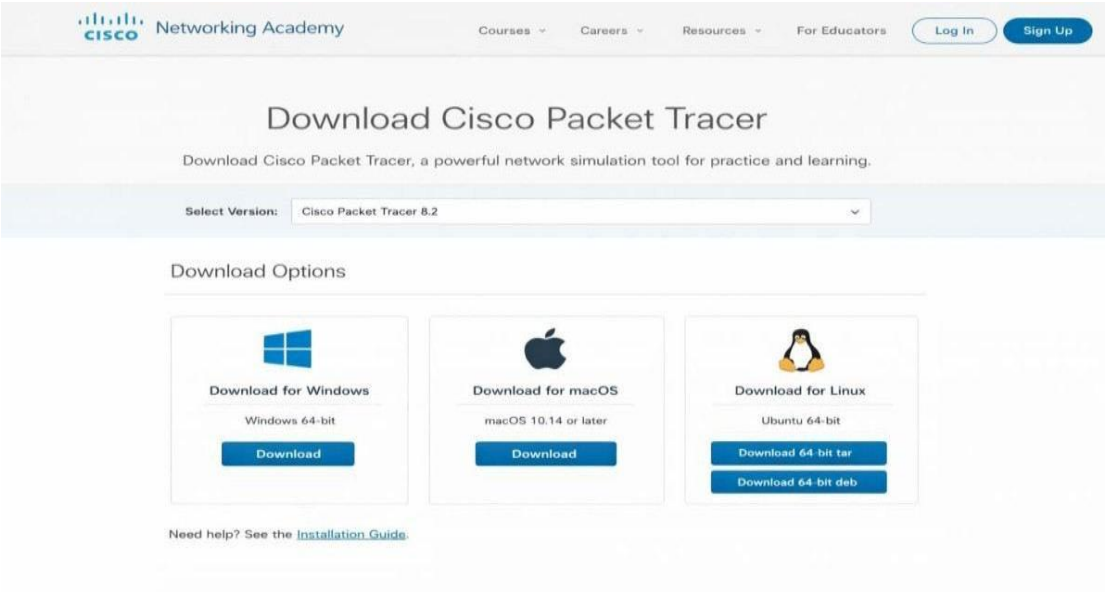


Figure 3: Cisco Packet Tracer Main Interface



Conclusion

Thus, Cisco Packet Tracer was successfully installed on the system. This experiment provided practical knowledge of installing and using network simulation software for learning and practice.

Experiment-08

Design various network Topology using packet tracer

Aim

To design and study different types of network topologies using Cisco Packet Tracer.

Hardware Used Software Used

- Computer system
- Cisco Packet Tracer
- Network devices (PCs, Switches, Routers in simulation)
- Ethernet cables (in Packet Tracer)

Theory

Introduction to Network Topology

Network topology refers to the arrangement of devices such as computers, switches, and routers in a network. It defines how devices are connected and how data flows between them. Understanding different topologies helps in designing efficient and reliable networks.

Types of Network Topologies

The main types of network topologies are:

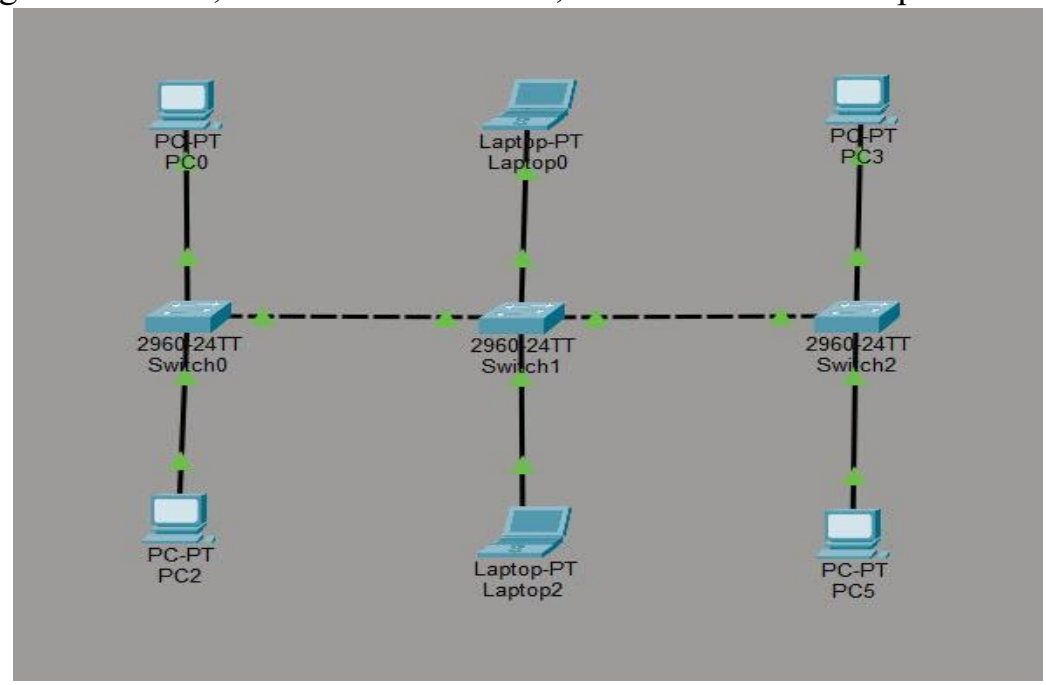
1. Bus Topology
2. Star Topology
3. Ring Topology
4. Mesh Topology
5. Tree Topology
6. Hybrid Topology

1. Bus Topology

Bus topology is one of the simplest network structures. In this topology, all devices are connected to a single main cable known as the **backbone cable**. Data sent from

a device travel along the backbone and is received by all devices, but only the intended device accepts the data.

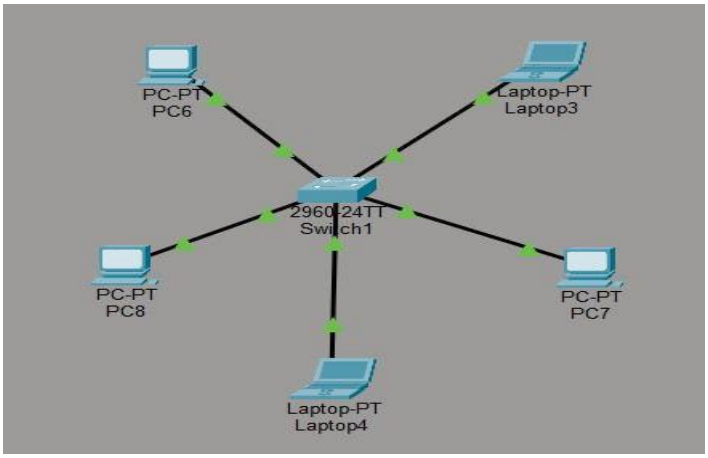
Bus topology is easy to install and requires less cable compared to other topologies. However, if the main cable fails, the entire network stops functioning.



2. Star Topology

In star topology, all devices are connected to a **central device** such as a hub or switch. All communication between devices passes through this central device.

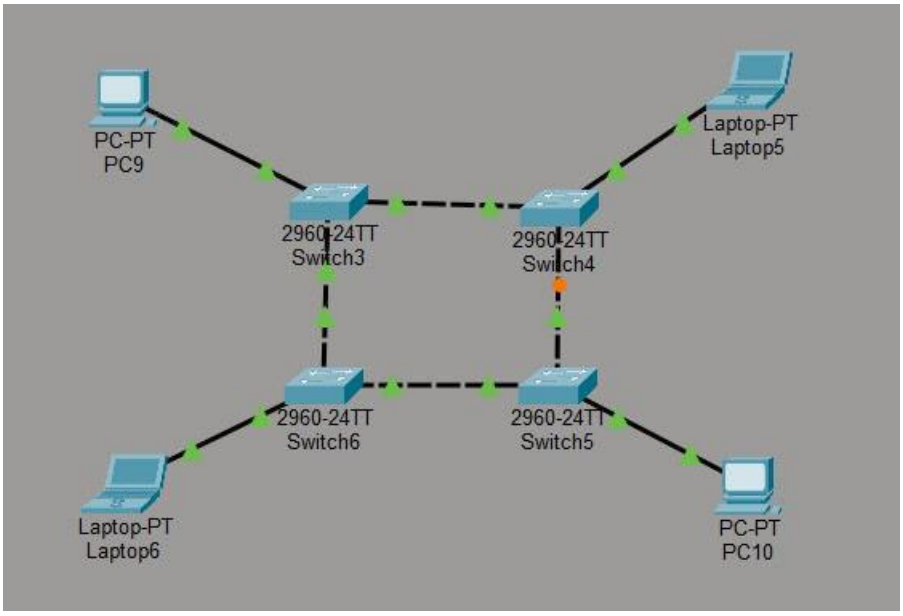
This topology is widely used in modern networks because it is easy to manage and troubleshoot. If one device fails, it does not affect the rest of the network. However,



if the central hub or switch fails, the entire network becomes inactive.

3. Ring Topology

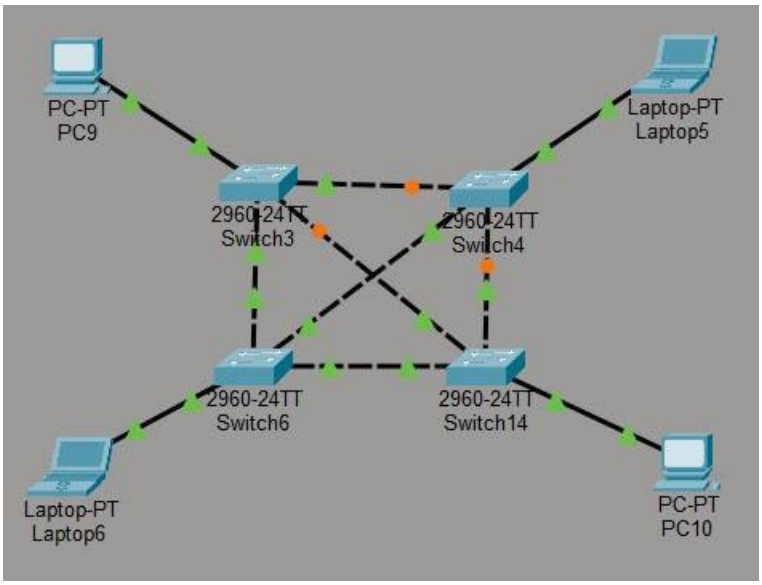
In ring topology, all devices are connected in a **circular loop**. Each device is connected to two other devices, forming a ring-like structure. Data travels in one direction from one device to another until it reaches the destination.
This topology reduces data collisions and ensures orderly data transmission.
However, if one device or connection fails, it may disrupt the entire network.



4.Mesh Topology

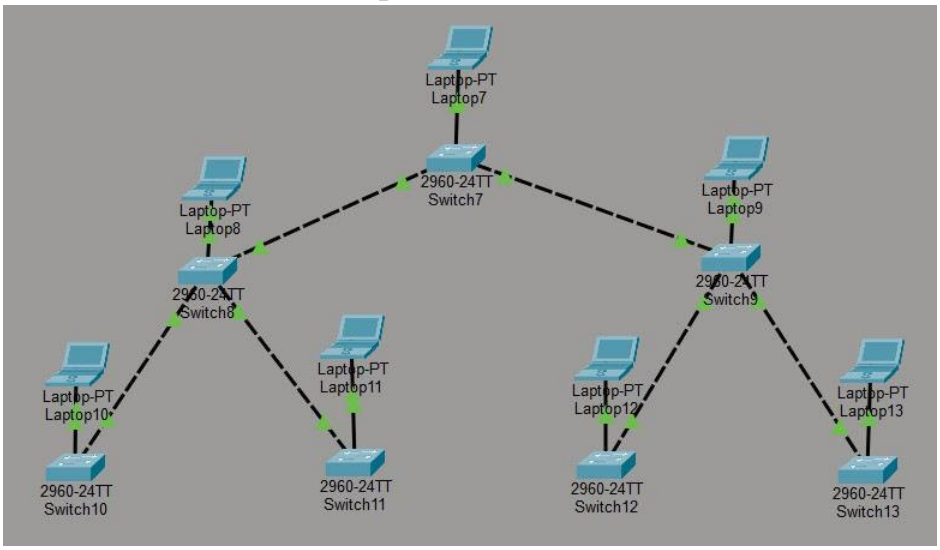
Mesh topology is a network structure in which every device is connected to every other device in the network. This creates multiple paths for data transmission.

Mesh topology provides high reliability and fault tolerance because if one link fails, data can be transmitted through another path. However, it requires a large number of cables and connections, making it expensive and complex.



4. Tree Topology

Tree topology is a hierarchical network structure that combines characteristics of **bus and star topologies**. In this topology, devices are arranged in a tree-like structure with a root node and multiple levels of nodes.



Tree topology is suitable for large networks because it allows easy expansion and management. However, if the main backbone cable fails, a large portion of the network may be affected.

6.Hybrid Topology

Hybrid topology is a combination of two or more different network topologies. For example, a network may combine star and bus topology to form a hybrid structure.

This topology provides flexibility and scalability, making it suitable for large organizations. However, hybrid networks are more complex to design and maintain.

Conclusion

In this experiment, various network topologies were designed and analyzed using Cisco Packet Tracer. By creating different topologies such as Bus, Star, Ring, Mesh, Tree, and Hybrid, we were able to understand how network devices are connected and how data is transmitted within a network. Each topology has its own structure, advantages, and limitations.

Through this practical work, we learned how network design affects performance, reliability, and scalability. The experiment also helped in gaining hands-on experience with Packet Tracer, which is useful for simulating and understanding real-world networking concepts. Therefore, studying and designing different network topologies is essential for building efficient and well-structured communication networks.

Experiment-10
Configuring IP addresses and Subnet mask using CISCO Packet Tracer

Aim
To configure IP addresses and subnet masks on network devices using Cisco Packet Tracer and verify network connectivity.

Hardware Used Software used
- Computer system
- Cisco Packet Tracer
- PCs and Switches (in simulation)
- Ethernet cables (in Packet Tracer)

Theory
Introduction to IP Addressing:
An IP address is a unique identifier assigned to each device in a network to enable communication. It helps in identifying the source and destination of data. There are two types of IP addressing:

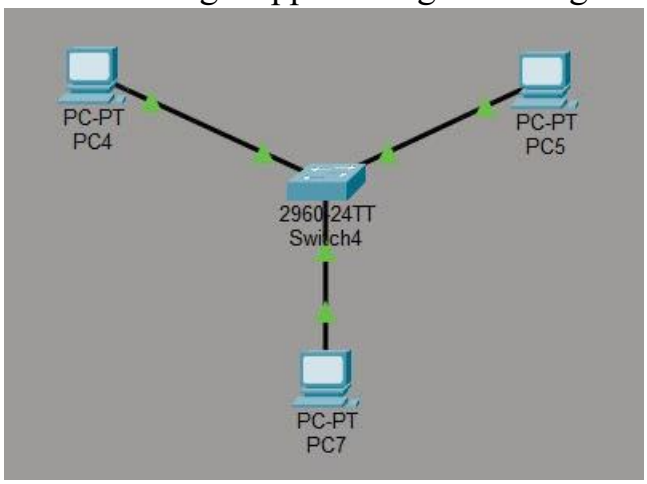
- **IPv4** (e.g., 192.168.1.1)
- **IPv6** (advanced version, larger address space)

Subnet Mask:
A subnet mask is used to divide an IP address into network and host portions. It helps devices determine whether a destination IP is in the same network or a different one.

Example:
- IP Address: 192.168.1.1
- Subnet Mask: 255.255.255.0

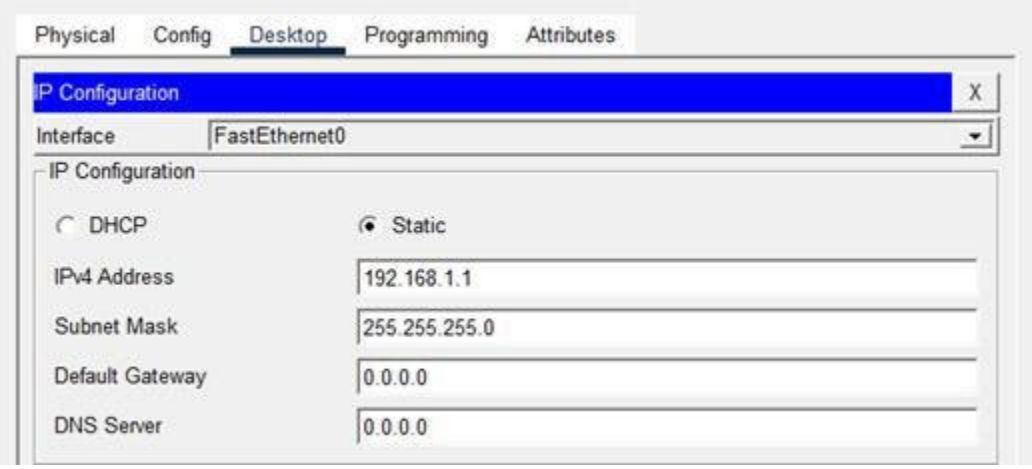
Steps to Configure IP Address in Packet Tracer
Step 1: Create Network Topology
- Open Cisco Packet Tracer

- Add two or more PCs and a Switch
- Connect devices using Copper Straight-Through cables.



Step 2: Assign IP Address and Subnet Mask

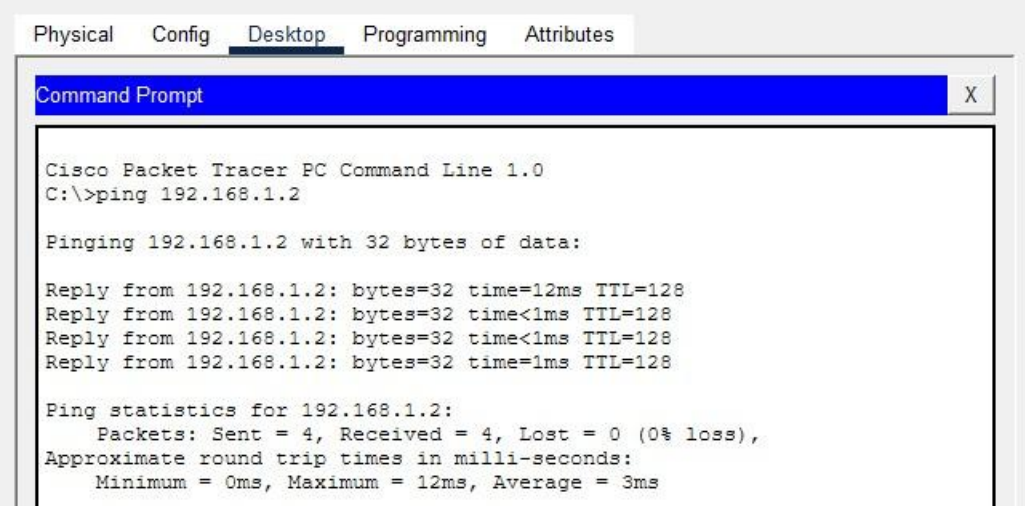
- Click on a PC → Go to *Desktop* → Select *IP Configuration*
- Enter the IP address and subnet mask manually Example:
- PC1:
  - IP Address: 192.168.1.1
  - Subnet Mask: 255.255.255.0
- PC2:
  - IP Address: 192.168.1.2
  - Subnet Mask: 255.255.255.0



Step 3: Verifying Network Connectivity

- Open *Command Prompt* from Desktop tab
- Use ping command to test connectivity Example:

ping 192.168.1.2



Conclusion

This experiment demonstrated how to configure IP addresses and subnet masks using Cisco Packet Tracer. Proper IP configuration is essential for communication between devices in a network and ensures efficient data transmission.

Experiment 11

Simulate TCP (Transmission Control Protocol) 3-Way Handshake using Wireshark

Aim

To study and simulate the TCP 3-way handshake process using Wireshark and analyze packet exchange.

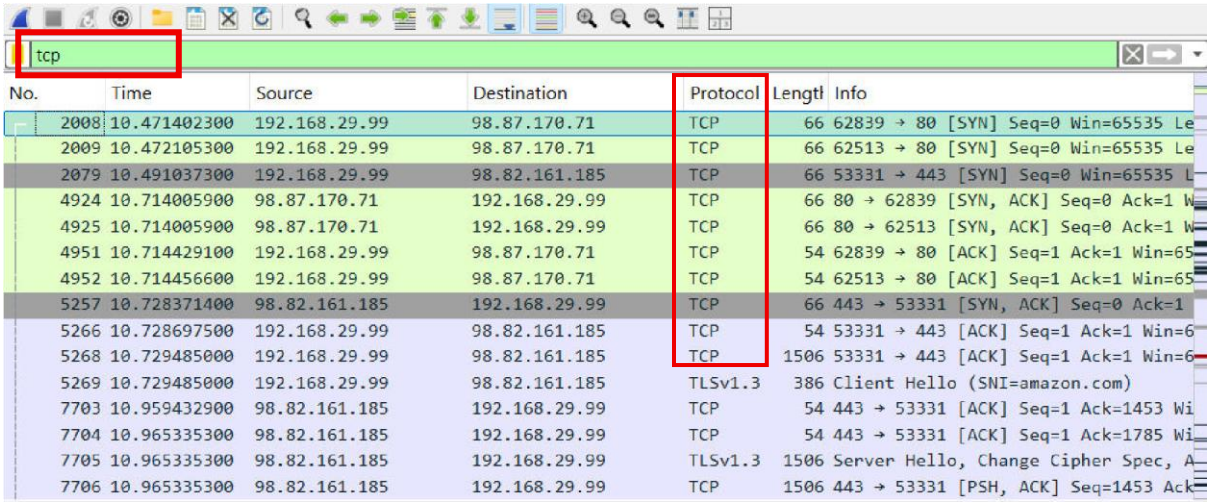
Hardware Used Software used

- Computer system
- Wireshark software
- Internet connection or local network
- Web browser / any network application

Theory

Introduction to TCP

Transmission Control Protocol (TCP) is a connection-oriented protocol used for reliable data transmission over a network. It ensures that data is delivered accurately, in order, and without loss.



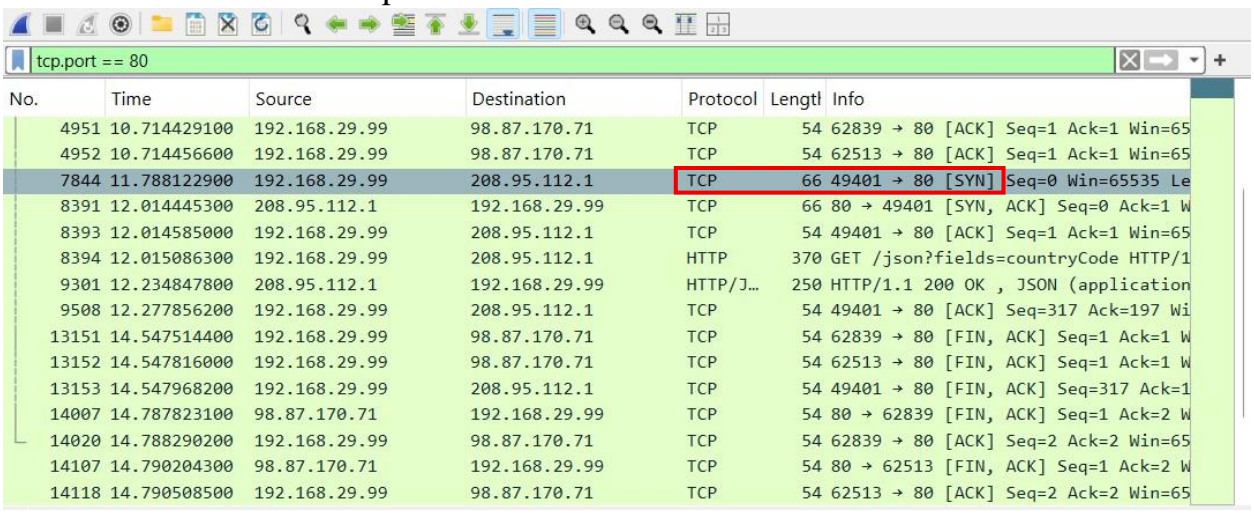
TCP 3-Way Handshake

The TCP 3-way handshake is a process used to establish a reliable connection between a client and a server before data transmission begins.

It involves three steps:

Step 1: SYN (Synchronize)

- The client sends a SYN packet to the server
- This indicates a request to establish a connection

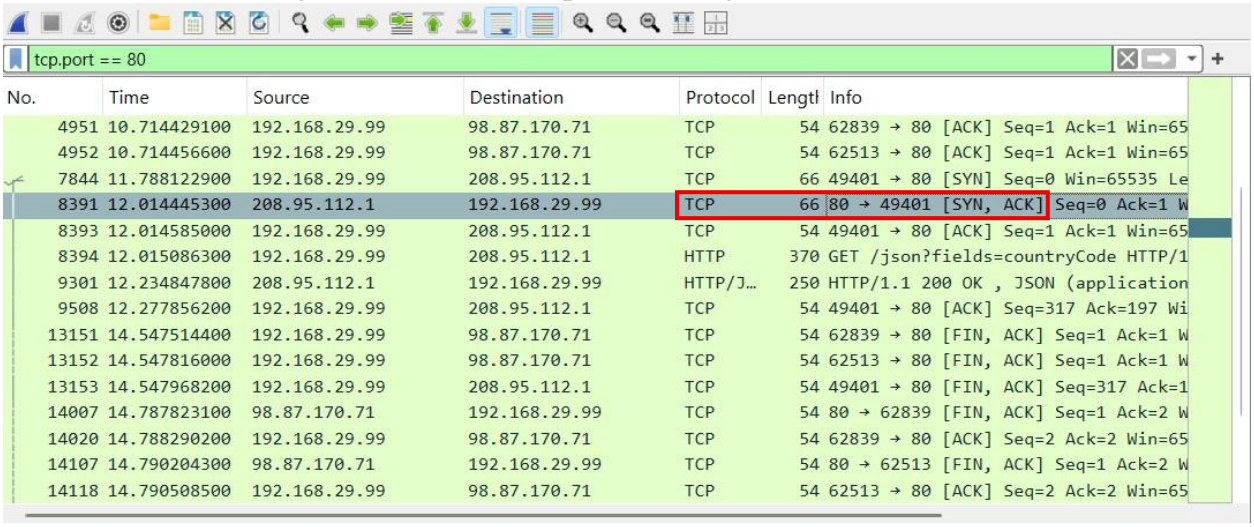


tcp.port == 80

No.	Time	Source	Destination	Protocol	Lengtl	Info
4951	10.714429100	192.168.29.99	98.87.170.71	TCP	54	62839 → 80 [ACK] Seq=1 Ack=1 Win=65
4952	10.714456600	192.168.29.99	98.87.170.71	TCP	54	62513 → 80 [ACK] Seq=1 Ack=1 Win=65
7844	11.788122900	192.168.29.99	208.95.112.1	TCP	66	49401 → 80 [SYN] Seq=0 Win=65535 Le
8391	12.014445300	208.95.112.1	192.168.29.99	TCP	66	80 → 49401 [SYN, ACK] Seq=0 Ack=1 W
8393	12.014585000	192.168.29.99	208.95.112.1	TCP	54	49401 → 80 [ACK] Seq=1 Ack=1 Win=65
8394	12.015086300	192.168.29.99	208.95.112.1	HTTP	370	GET /json?fields=countryCode HTTP/1
9301	12.234847800	208.95.112.1	192.168.29.99	HTTP/J…	250	HTTP/1.1 200 OK , JSON (application
9508	12.277856200	192.168.29.99	208.95.112.1	TCP	54	49401 → 80 [ACK] Seq=317 Ack=197 Wi
13151	14.547514400	192.168.29.99	98.87.170.71	TCP	54	62839 → 80 [FIN, ACK] Seq=1 Ack=1 W
13152	14.547816000	192.168.29.99	98.87.170.71	TCP	54	62513 → 80 [FIN, ACK] Seq=1 Ack=1 W
13153	14.547968200	192.168.29.99	208.95.112.1	TCP	54	49401 → 80 [FIN, ACK] Seq=317 Ack=1
14007	14.787823100	98.87.170.71	192.168.29.99	TCP	54	80 → 62839 [FIN, ACK] Seq=1 Ack=2 W
14020	14.788290200	192.168.29.99	98.87.170.71	TCP	54	62839 → 80 [ACK] Seq=2 Ack=2 Win=65
14107	14.790204300	98.87.170.71	192.168.29.99	TCP	54	80 → 62513 [FIN, ACK] Seq=1 Ack=2 W
14118	14.790508500	192.168.29.99	98.87.170.71	TCP	54	62513 → 80 [ACK] Seq=2 Ack=2 Win=65

Step 2: SYN-ACK (Synchronize Acknowledge)

- The server responds with a SYN-ACK packet
- It acknowledges the client's request and agrees to establish the connection

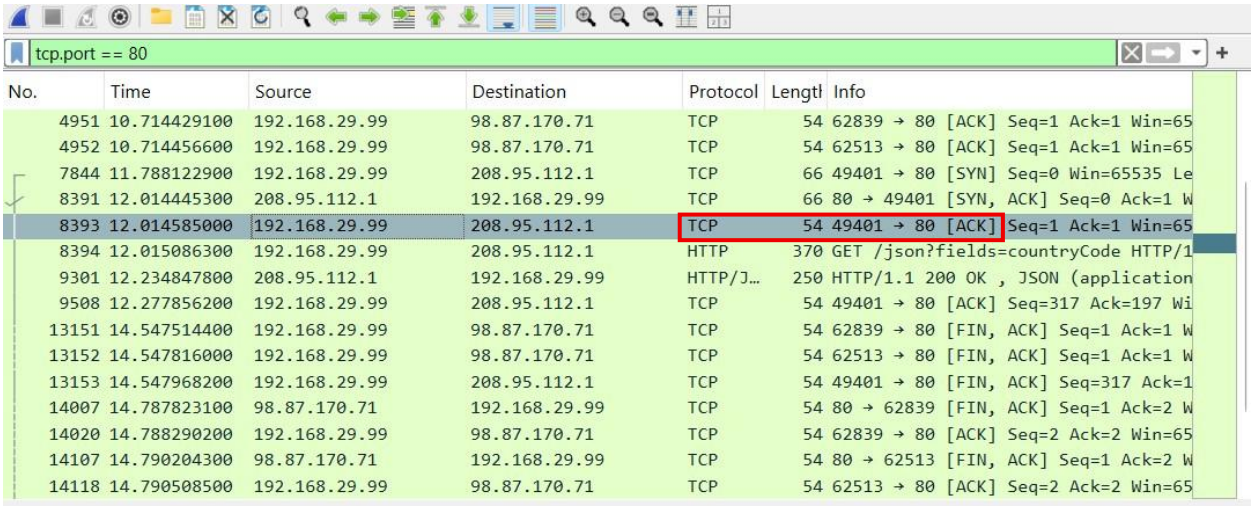


tcp.port == 80

No.	Time	Source	Destination	Protocol	Lengtl	Info
4951	10.714429100	192.168.29.99	98.87.170.71	TCP	54	62839 → 80 [ACK] Seq=1 Ack=1 Win=65
4952	10.714456600	192.168.29.99	98.87.170.71	TCP	54	62513 → 80 [ACK] Seq=1 Ack=1 Win=65
7844	11.788122900	192.168.29.99	208.95.112.1	TCP	66	49401 → 80 [SYN] Seq=0 Win=65535 Le
8391	12.014445300	208.95.112.1	192.168.29.99	TCP	66	80 → 49401 [SYN, ACK] Seq=0 Ack=1 W
8393	12.014585000	192.168.29.99	208.95.112.1	TCP	54	49401 → 80 [ACK] Seq=1 Ack=1 Win=65
8394	12.015086300	192.168.29.99	208.95.112.1	HTTP	370	GET /json?fields=countryCode HTTP/1
9301	12.234847800	208.95.112.1	192.168.29.99	HTTP/J…	250	HTTP/1.1 200 OK , JSON (application
9508	12.277856200	192.168.29.99	208.95.112.1	TCP	54	49401 → 80 [ACK] Seq=317 Ack=197 Wi
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13152	14.547816000	192.168.29.99	98.87.170.71	TCP	54	62513 → 80 [FIN, ACK] Seq=1 Ack=1 W
13153	14.547968200	192.168.29.99	208.95.112.1	TCP	54	49401 → 80 [FIN, ACK] Seq=317 Ack=1
14007	14.787823100	98.87.170.71	192.168.29.99	TCP	54	80 → 62839 [FIN, ACK] Seq=1 Ack=2 W
14020	14.788290200	192.168.29.99	98.87.170.71	TCP	54	62839 → 80 [ACK] Seq=2 Ack=2 Win=65
14107	14.790204300	98.87.170.71	192.168.29.99	TCP	54	80 → 62513 [FIN, ACK] Seq=1 Ack=2 W
14118	14.790508500	192.168.29.99	98.87.170.71	TCP	54	62513 → 80 [ACK] Seq=2 Ack=2 Win=65

Step 3: ACK (Acknowledge)

- The client sends an ACK packet back to the server
- Connection is successfully established



Steps to Perform in Wireshark

Step 1: Start Packet Capture

- Open Wireshark
- Select the active network interface (Wi-Fi/Ethernet)
- Start capturing packets

Step 2: Generate Traffic

- Open a web browser
- Visit any website (e.g., google.com) to generate TCP traffic

Step 3: Apply Filter

- Apply filter: tcp/tcp.port == 80
- Identify handshake packets (SYN, SYN-ACK, ACK)

Step 4: Analyze Packets

- Click on packets to view details
- Observe flags in TCP header
- Identify the sequence of SYN → SYN-ACK → ACK

Conclusion

This experiment demonstrated the TCP 3-way handshake using Wireshark. It helped in understanding how a reliable connection is established between client and server before data transmission begins.

Experiment 13
Configure Various Networking Devices Using Cisco Packet Tracer

Aim

To configure different networking devices such as PCs, switches, and routers using Cisco Packet Tracer.

Hardware Used Software Used

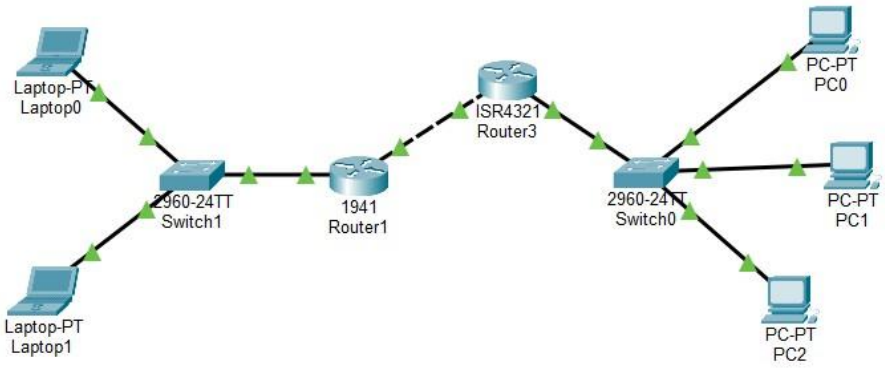
- Computer system
- Cisco Packet Tracer
- Network devices (PCs, Switches, Routers in simulation)
- Ethernet cables (in Packet Tracer)

Theory

Introduction to Networking Devices

Networking devices are used to connect multiple computers and enable communication within and between networks. Each device has a specific role in data transmission.

- **PC (End Device):** Used to send and receive data
- **Switch:** Connects multiple devices within a LAN
- **Router:** Connects different networks and routes data



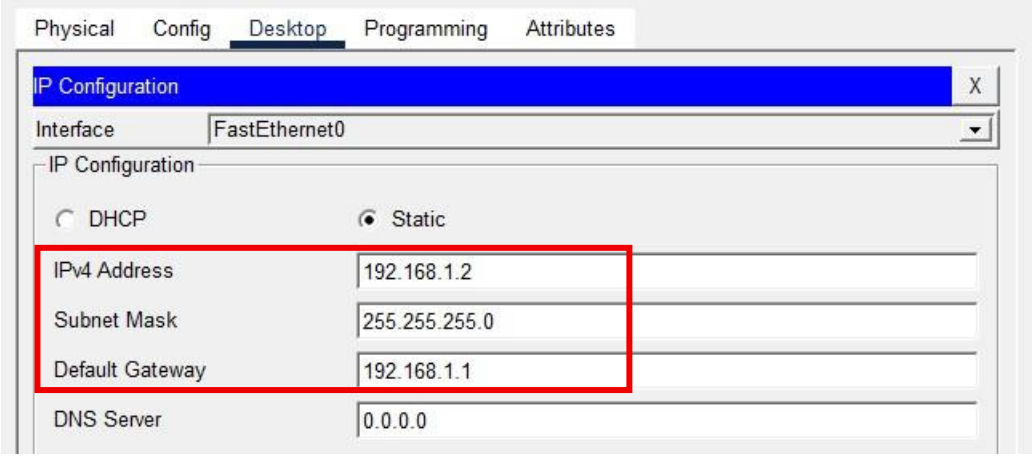
Basic Configuration of Devices in Packet Tracer

1. PC Configuration

PCs are configured by assigning IP addresses so that they can communicate within a network.

Steps:

- Click on PC → Go to *Desktop* → Select *IP Configuration*
- Enter IP address and subnet mask Example:
- IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1



2. Switch Configuration

Switches are used to connect devices in a LAN. Basic configuration includes assigning a hostname and managing ports.

Steps:

- Click on Switch → Go to *CLI* tab Enter the following commands:
  - Enable
  - configure terminal
  - hostname Switch1
- (Optional) Assign IP to VLAN interface: - interface vlan 1
  - IP address 192.168.1.10 255.255.255.0
  - no shutdown

```
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2,
changed state to up


Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Switch1
Switch1(config)#interface vlan 1
Switch1(config-if)#ip address 192.168.1.10 255.255.255.0
Switch1(config-if)#no shutdown

Switch1(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

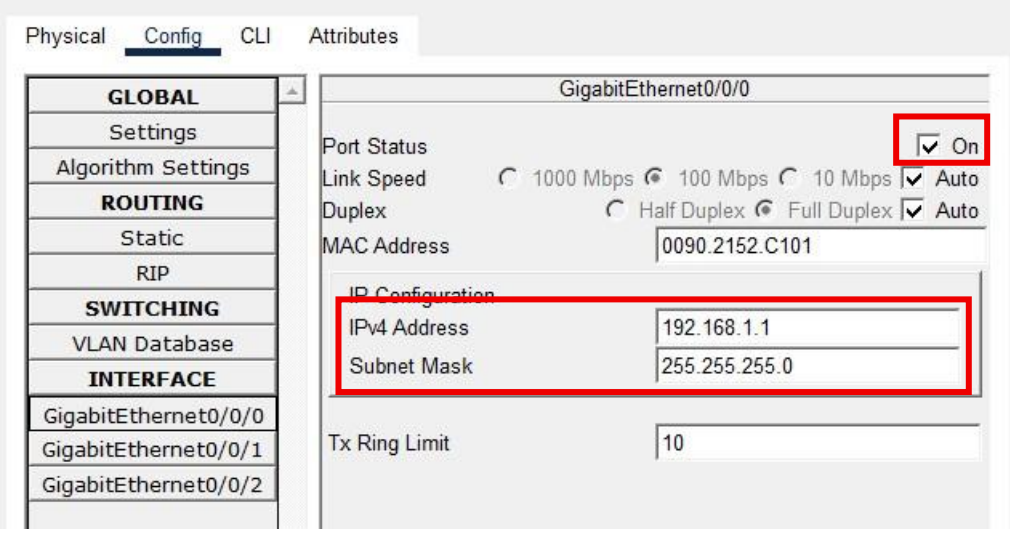
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to
up
```

3. Router Configuration

Routers are configured to connect different networks and enable communication between them.

Steps:

- Click on Router → Go to *config* tab
- Under 'Interface' select 'gigabitEthernet 0/0'
- Check the port status to 'on'
- Enter the IPv4 address: 192.168.1.1
- Enter subnet mast: 255.255.255.0

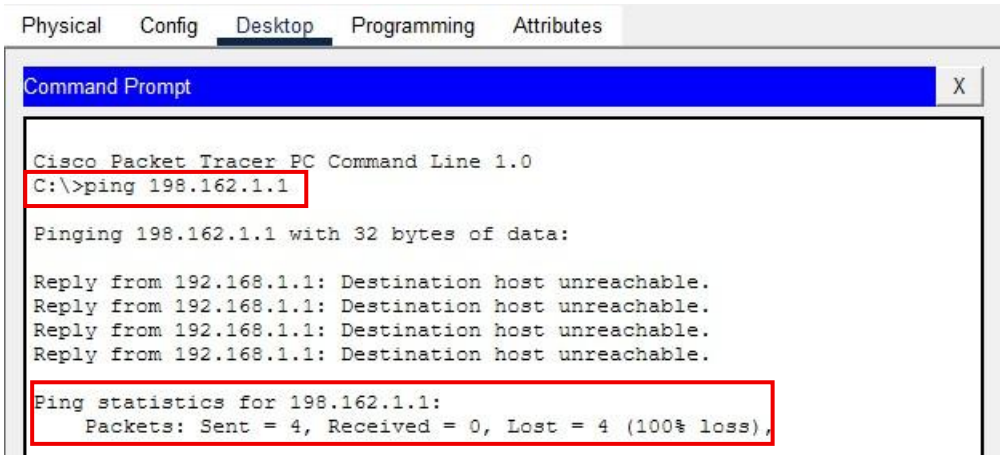


4. Verifying Configuration

- Use ping command from PC to test connectivity

- Check if devices can communicate successfully Example:

ping 192.168.1.1



Conclusion

This experiment demonstrated how to configure various networking devices such as PCs, switches, and routers using Cisco Packet Tracer. Proper configuration ensures effective communication and network functionality.

Experiment 14

Study of Application Layer Protocols – DNS, FTP, SMTP, SNMP and HTTP

Aim

To study various application layer protocols such as DNS, FTP, SMTP, SNMP, and HTTP and understand their functions in networking.

Hardware Used Software Used

- Computer system
- Web browser

Theory

Introduction to Application Layer

The application layer is the topmost layer of the OSI model and acts as an interface between the user and the network. It provides network services directly to end-user applications such as web browsers, email clients, and file transfer utilities. Application layer protocols define how data is formatted, transmitted, and received between applications over a network.

1. DNS (Domain Name System) Definition:

DNS is a hierarchical and distributed naming system used to translate humanreadable domain names into IP addresses.

Working:

When a user enters a domain name in a browser, a DNS query is sent to a DNS server, which resolves the domain name into its corresponding IP address. This process may involve multiple DNS servers such as root, TLD, and authoritative servers.

Function:

- Converts human-readable domain names into machine-readable IP addresses
- Helps users access websites easily

- Reduces the need to remember IP addresses **Example:**

Typing www.google.com in a browser gets converted into an IP address

2. FTP (File Transfer Protocol) Definition:

FTP is a standard protocol used for transferring files between a client and a server over a TCP/IP network.

Working:

FTP uses two connections:

- **Control connection** (port 21) for commands
- **Data connection** (port 20) for file transfer

Function:

- Uploading and downloading files
- Managing files on a remote server
- Supports authentication (username/password) **Example:**

Uploading a website to a web server

3. SMTP (Simple Mail Transfer Protocol) Definition:

SMTP is a protocol used for sending emails from a client to a mail server and between mail servers.

Working:

When an email is sent, SMTP transfers the message from the sender's mail server to the recipient's mail server. It works on a store-and-forward mechanism.

Function:

- Handles outgoing mail
- Transfers email messages between servers
- Works with protocols like POP3/IMAP for receiving emails **Example:**

Sending an email using Gmail

4. SNMP (Simple Network Management Protocol) Definition:

SNMP is an application layer protocol used for monitoring and managing network devices such as routers, switches, and servers.

Working:
SNMP operates using a manager-agent model:

- **Manager:** Central system that monitors devices
- **Agent:** Software on devices that collects and sends data

Function:

- Collects information about network performance
- Helps in network management and troubleshooting
- Configuration management

Example:
Monitoring network traffic on a router

5. HTTP (HyperText Transfer Protocol)
Definition:
HTTP is a protocol used for transferring web pages and resources between a client (browser) and a web server.

Working:
HTTP follows a request-response model:

- Client sends an HTTP request
- Server responds with requested data

Function:

- Enables browsing of websites
- Transfers text, images, and other web content
- Stateless protocol (each request is independent)

Example:
Accessing a website using a browser

Conclusion

This experiment provided a detailed understanding of key application layer protocols including DNS, FTP, SMTP, SNMP, and HTTP. These protocols enable essential services such as web browsing, email communication, file transfer, and network management, forming the backbone of modern networking.